Department of Electrical and Infromation Engineering

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Threat Containment Strategies for Compromised xApps in Open RAN

Undergraduate Project Proposal

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Introduction

The foundational transition from traditional RAN systems to the **O-RAN**, **Problem Statement**, **Objectives and Scope** of the proposed threat containment framework

Background

The Shift to Open RAN

- **Evolution of the RAN**: Transition from **monolithic, vendor-locked proprietary hardware** to **open**, **disaggregated**, and **software-based (virtualized) components**.
- **Architectural Disaggregation**: Breaking the **5G base station** (gNB) into three logical units as: **O-CU** (O-RAN Central Unit), **O-DU** (O-RAN Distributed Unit) and **O-RU** (O-RAN Radio Unit).
- **Open Interfaces**: Integration of multi-vendor components facilitated by **standardized interfaces like E2, A1, O1 and O2**.

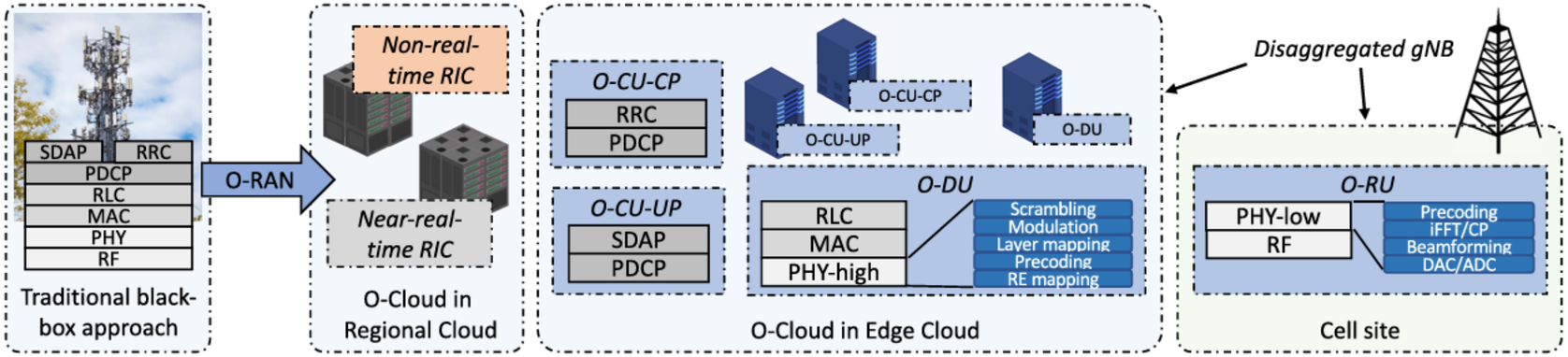


Figure 1: Evolution of traditional base station to a virtualized gNB with dissagregation [1]

Background

Open RAN Architecture

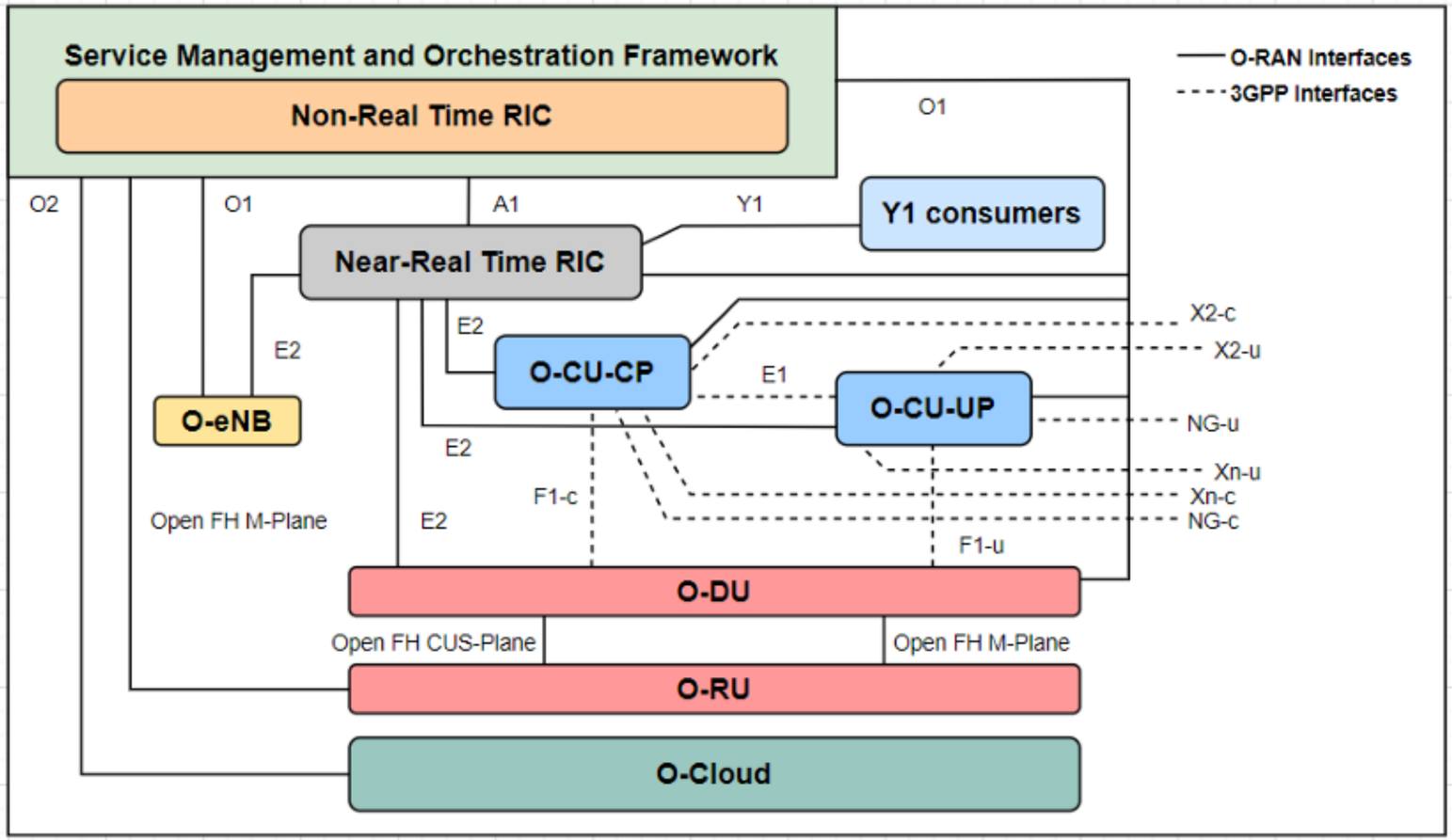


Figure 2: Logical architecture of Open RAN [2]

Background

Near-RT RIC and xApps

- **Near-Real-Time RIC**: A microservice-based platform that **hosts intelligent control applications** with control loops between **10ms and 1s**.
- **Role of xApps**: Third-party applications that perform critical tasks like **near-real-time RAN control and optimization** etc.
- The **E2 Interface**: The critical data pipe used by xApps to **collect RAN measurements and issue control commands to E2 nodes**.

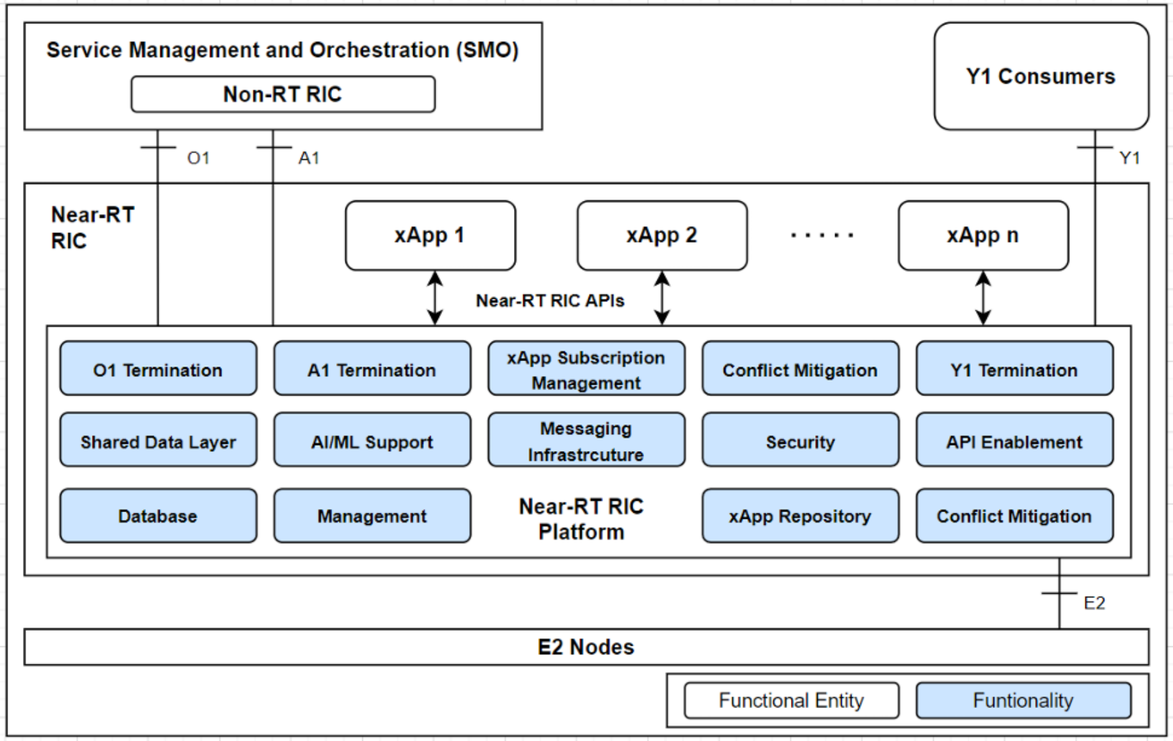


Figure 3: Near-RT RIC Architecture [3]

Problem Statement

Compromised xApps in Open RAN

- **Expanded Attack Surface**: O-RAN design allows third party **xApps** to **execute control actions within the Near Real Time RIC**.
- A **compromised xApp** may:
  - Misuse shared resources, interfere with other xApps, move laterally within the RIC environment and negatively impact RAN performance.
  - Misuse user data which can cause privacy issues and manipulate network functions.
- **Workload identity attestation** of xApps, **continuous authentication** and **runtime isolation of compromised xApps** are not yet well considered.
- This creates a critical security gap in multi-vendor Open RAN environments which this project aims to address.

Objectives & Scope

- **Primary Objective**: Implement a **security framework** that enhances the **resilience of Near-RT RIC against threats from compromised xApps by isolating them**.

- **Scope of the project**:
  - Implement a **remote attestation** framework for integrity verification of xApp infrastructure.
  - Implement a **workload identity attestation** mechanism for identity and integrity verification of xApps.
  - Develop an **isolation pipeline** that is triggered by security violations of xApps to **contain compromised xApps**.
  - Experimentally validate the proposed framework within a testbed.

Literature Review

Analyzing architectural vulnerabilities and current **security limitations** to justify the transition toward automated, **real-time** threat containment.

The Evolution of the Threat Landscape

- Shift from "**Black-Box**" to Disaggregation: Traditional RAN relied on "**security by obscurity**" within a single vendor perimeter.

- Dissolution of Trust: In O-RAN, components from multiple vendors must communicate, meaning physical location no longer implies legitimacy.

- The E2 Attack Vector: The critical artery between the RIC and RAN nodes is susceptible to **Man-in-the-Middle (MitM)** attacks and **signaling floods**.

- **Supply Chain Risks**: Third-party xApps introduce vulnerabilities through untrusted code or compromised libraries.

Impacts of Compromised xApps

RESOURCE EXHAUSTION (DOS)

Malicious xApps can starve neighboring xApps of CPU/memory, crippling network performance.

DATA EXFILTRATION

Breaching the Shared Data Layer (SDL) allows theft of sensitive User Equipment (UE) identities like IMSI.

CONTROL LOOP ABUSE

Malicious policies can manipulate radio resources to track user locations or degrade specific services.

PROVEN VULNERABILITIES

Real world examples like CVE-2023-42358 show that a single malformed packet can crash the E2 Termination.

Critical Gaps & the Need for Automatic Containment

STATIC VS. DYNAMIC DEFENSE

Current security relies on pre-deployment scanning; it cannot stop novel attacks during runtime.

THE "HUMAN BOTTLENECK"

Most current frameworks require human intervention to analyze logs and terminate pods. [6]

LATENCY-SECURITY TRADE OFF

There is a lack of empirical data on whether continuous Zero Trust checks will violate 5G's sub-second timing .

Methodology

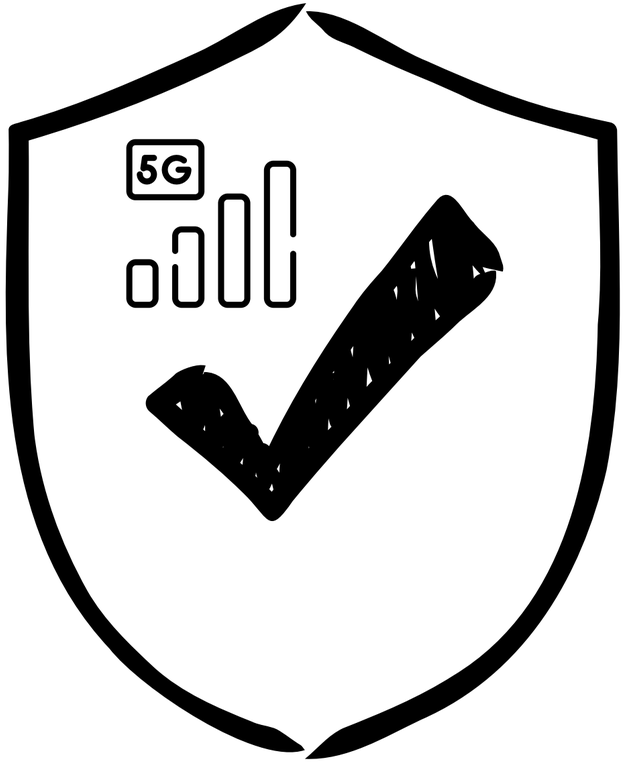
Proposed workflow: **Zero Trust**–based security framework for Near-RT RIC. Focus on compromised **xApp containment**, not only prevention.

Frameworks and Organizations

- srsRAN Project [4]
- OSC [5]
- Open-Source Software Tools
- TCG

Proposed Structure

- Hardware Trust
  - Remote Attestation
- Software Trust
  - Workload Identity Attestation
  - Runtime Monitoring
  - Automated Isolation



01. Remote Attestation

Mechanism: RoT | TPM | TEE | PCR

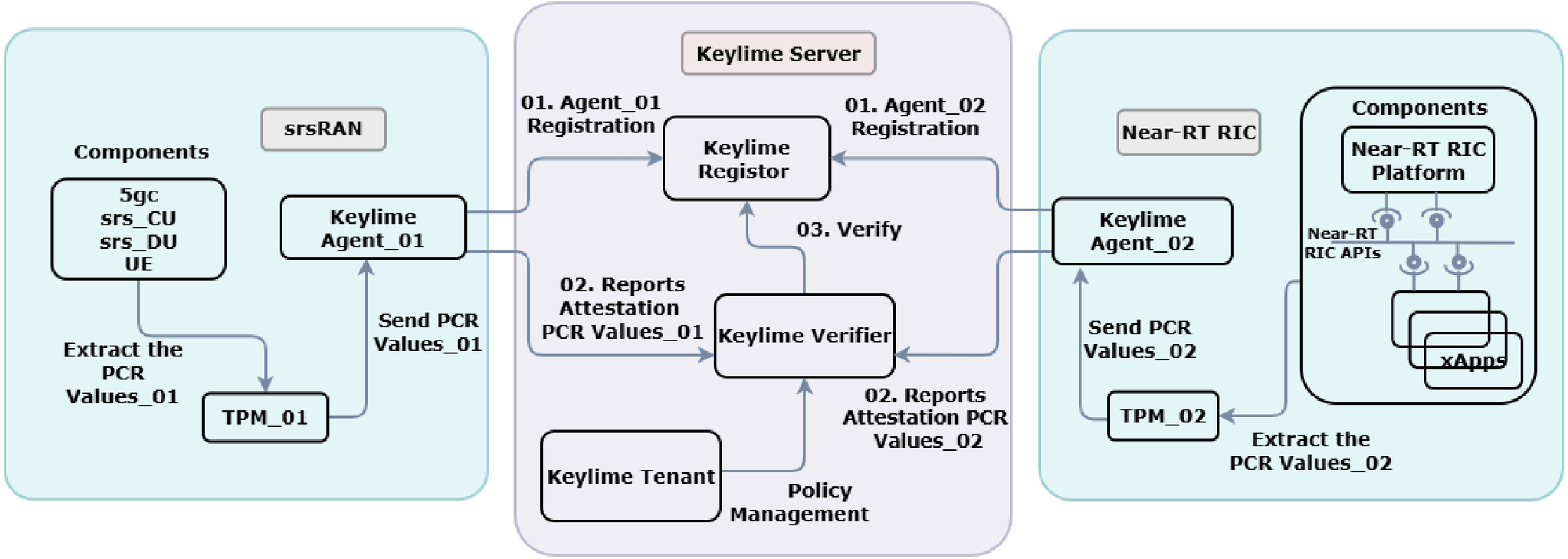


Figure 4: Measurement-Based Remote Attestation Workflow

02. Workload Attestation and Isolation

Mechanism:

- Workload Identity Attestation
  SPIRE < Agent / Server
- xApp Behavior Monitoring - cAdvisor, falco
- Isolation

Mechanism: Container-Level Isolation | Docker cgroups | Seccomp Profiles

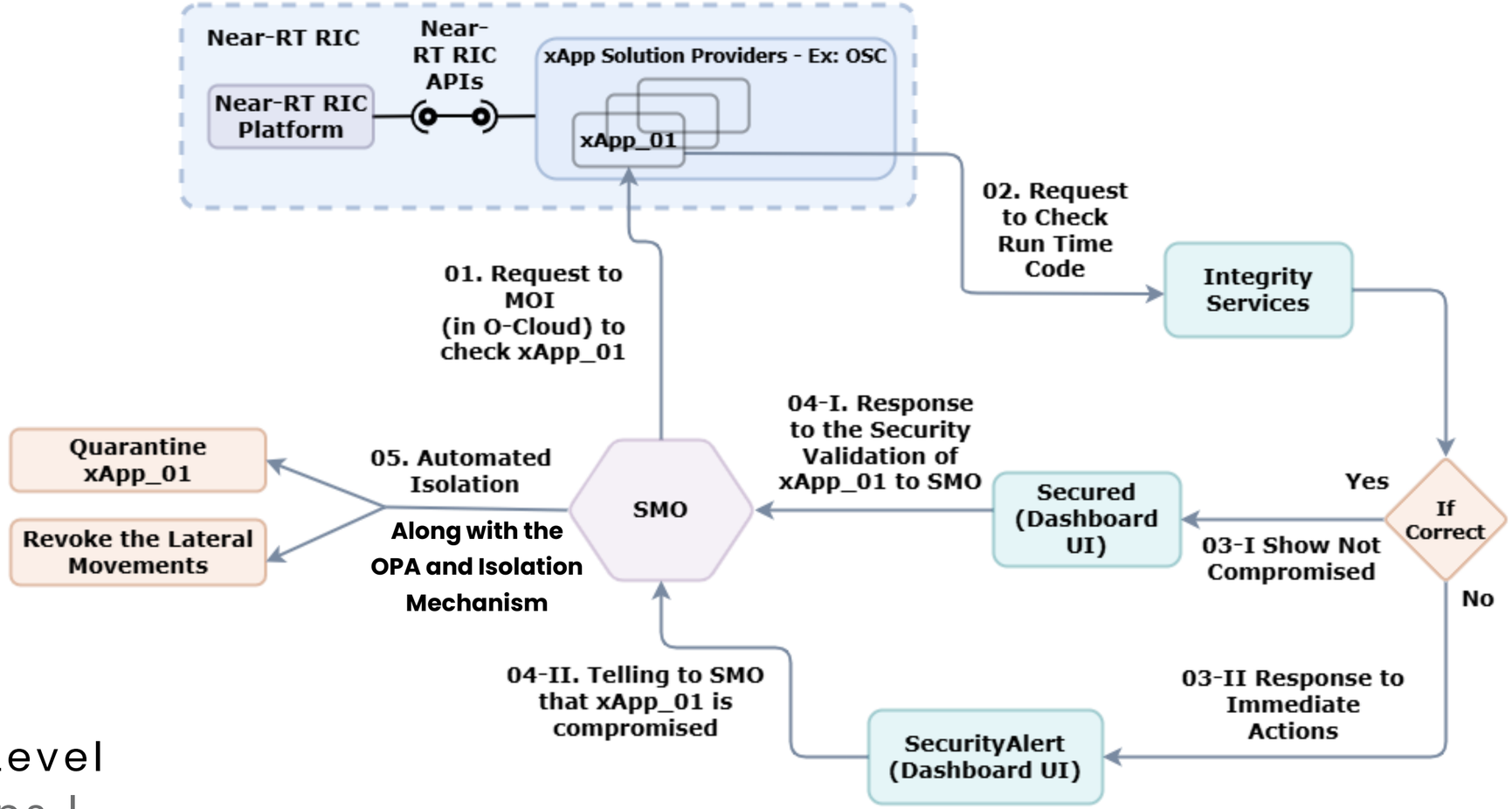
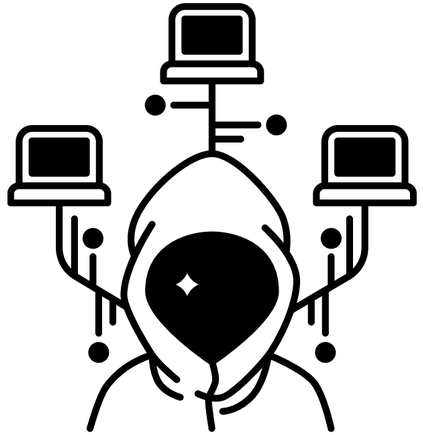
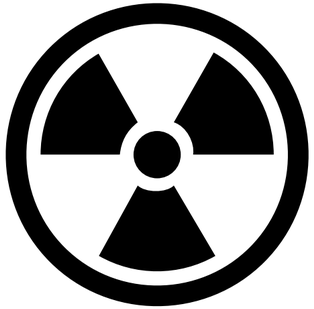


Figure 5: xApp Validation Procedure and Isolation

Testing and Evaluation



- Integrity of Architecture
- Simulation Attacks
- Evaluation of the Performance Metrics
- Isolated xApp Identification



Resources Required & Timeline Management

HARDWARE REQUIREMENTS

- Computers capable of running virtual machines and containers
- Virtual machines for flexible deployment and scalability
- Optional Raspberry Pi for lightweight simulations (e.g., SMO)
- Trusted Platform Module (TPM) for remote attestation
- User Equipment (UE) for connectivity testing

SOFTWARE REQUIREMENTS

- Ubuntu Linux as the operating system
- O-RAN and srsRAN software frameworks for testbed setup
- Keylime for remote attestation
- Workload integrity (e.g., SPIRE)
- xApp verification platform (e.g., cAdvisor, Falco)
- Docker and Kubernetes for containerization and isolation
- User interface to monitor attestation and xApp integrity

Timeline

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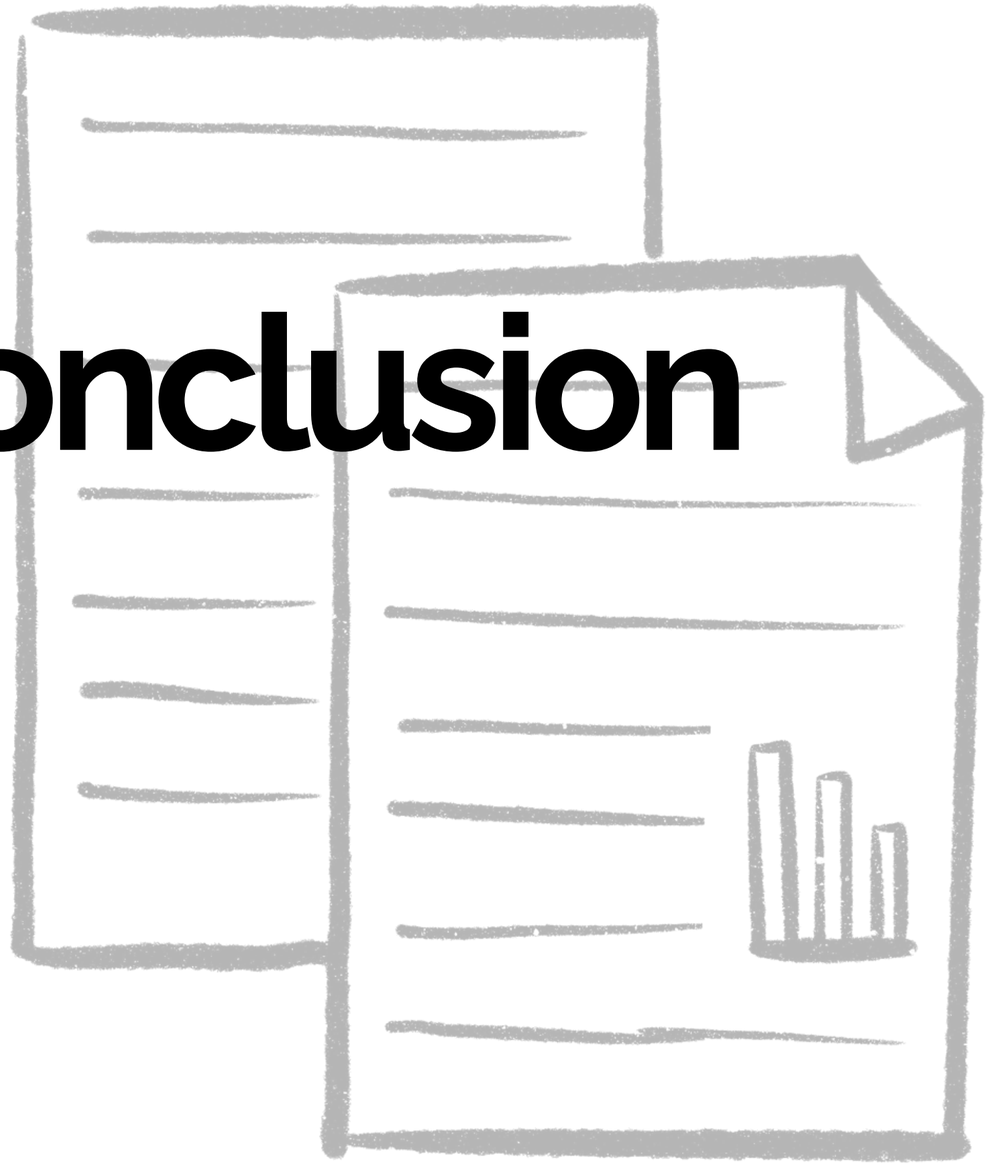
Task	2025/2026																																							
	Nov				Dec				Jan				Feb				Mar				Apr				May				Jun				Jul				Aug			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Selection of the Project Topic	■	■																																						
Discussion with Supervisor and Co-Supervisors			■																																					
Study on Research Work (Papers, Journals, Conferences)				■	■	■	■	■																																
Prepare Project Proposal and Presentation									■	■																														
Study on Existing O-RAN Test Frameworks										■	■																													
Initial Testbed Setup and Design											■	■																												
Remote Attestation using TPM													■	■	■	■																								
Design xApp Integrity and Workload Attestation																■	■	■	■	■	■																			
Quarantine and Isolation of Compromised xApps																					■	■	■	■	■	■	■	■	■											
Integration Demonstration and Visualization																														■	■	■	■							
Testing and Evaluation																																	■	■	■	■				
Final Report Submission and Presentation																																					■	■	■	■

Conclusion



Conclusion

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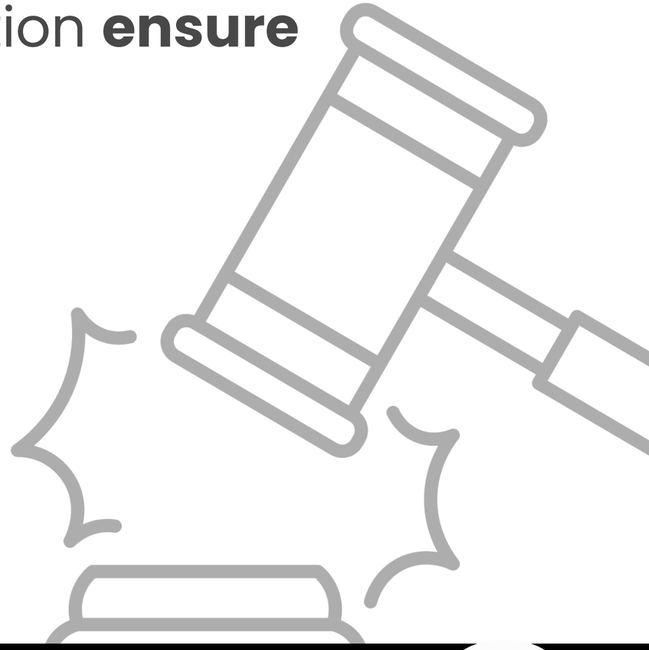
- O-RAN increases flexibility but also introduces new security risks through third-party xApps
- This project proposes a zero-trust threat containment framework for Near-RT RIC
- **Continuous remote attestation** and workload identity verification **ensure runtime trust**
- Automatic isolation contains compromised xApps effectively



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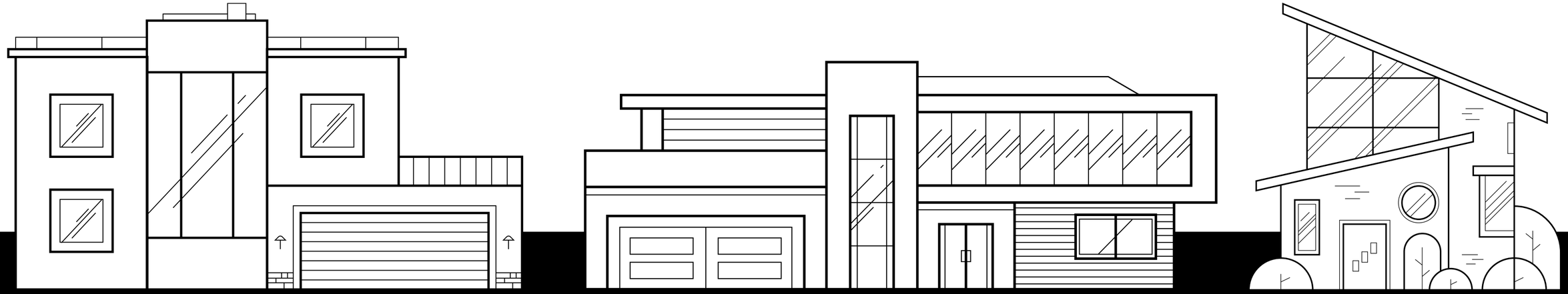
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